Private, Secure, and Cost-Effective
The Trifecta of Managed DevOps Pools

Wolfgang Ofner
Senior Cloud Architect

Agenda

Azure Security

Azure DevOps Self-Hosted Agents

Managed DevOps Pools

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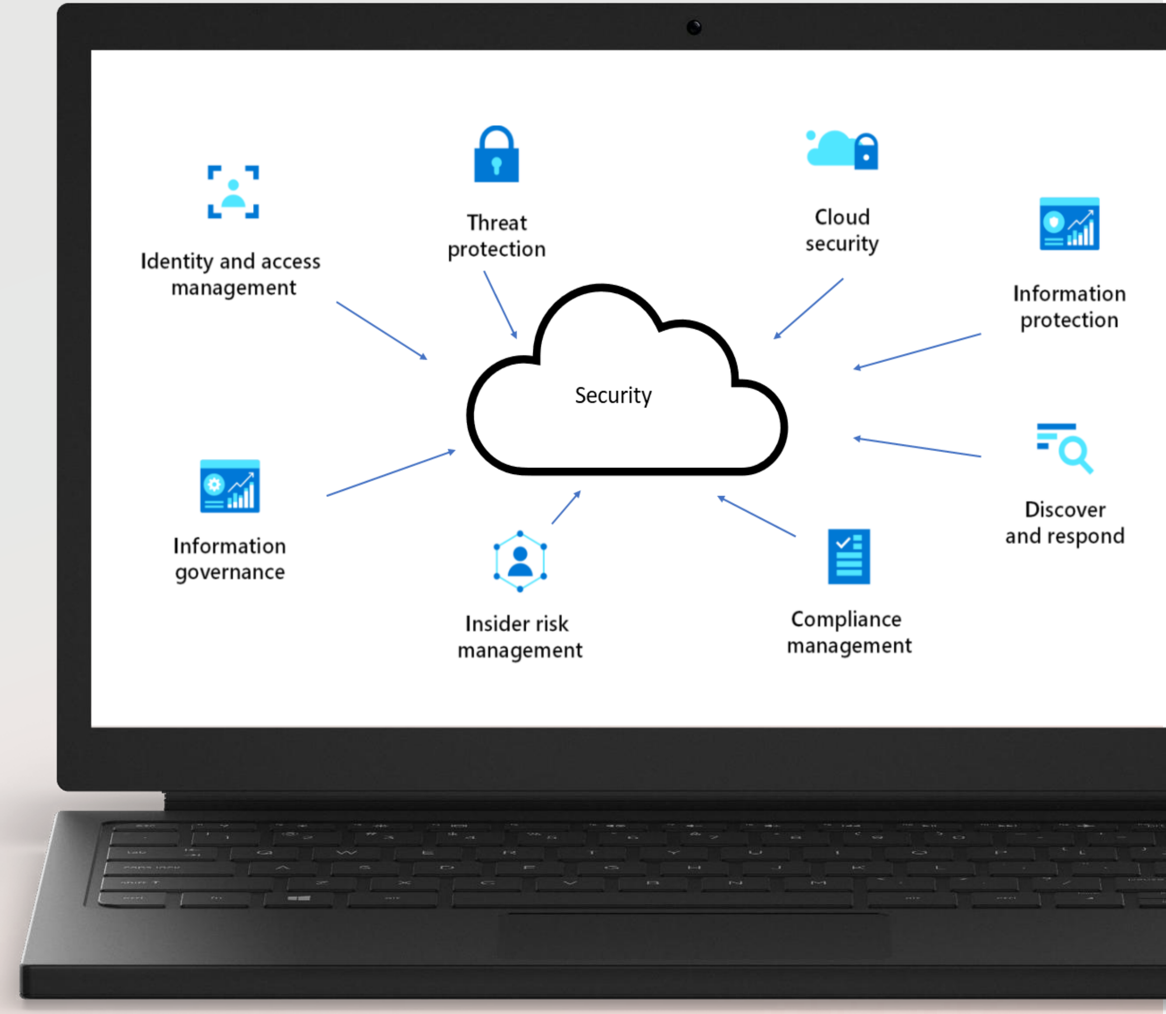
Azure Security

Azure Security

Passwords and secrets

Azure services are available over the internet

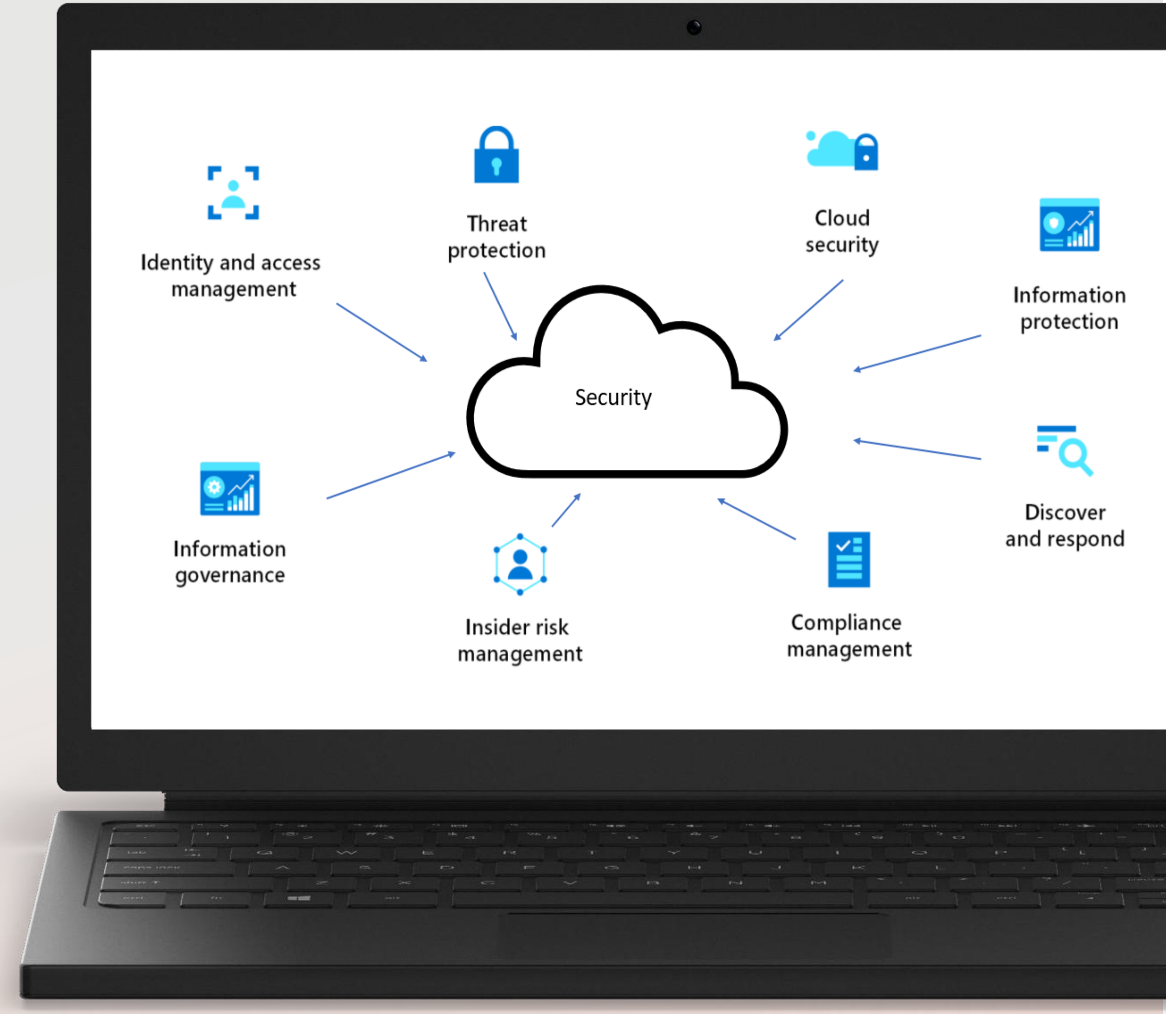


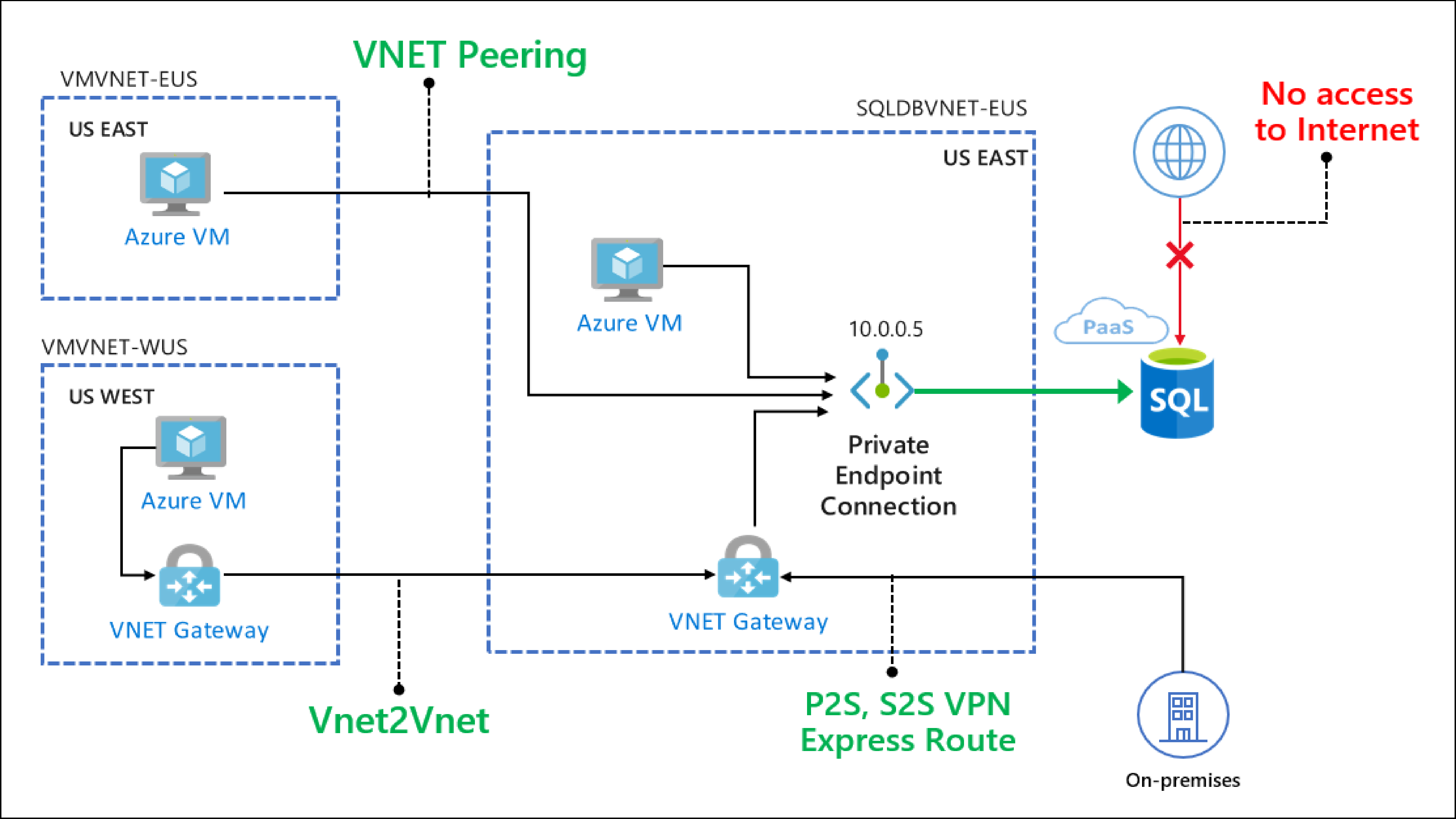
Azure Security

Managed Identities

Only allow access from within VNet

Private Endpoints





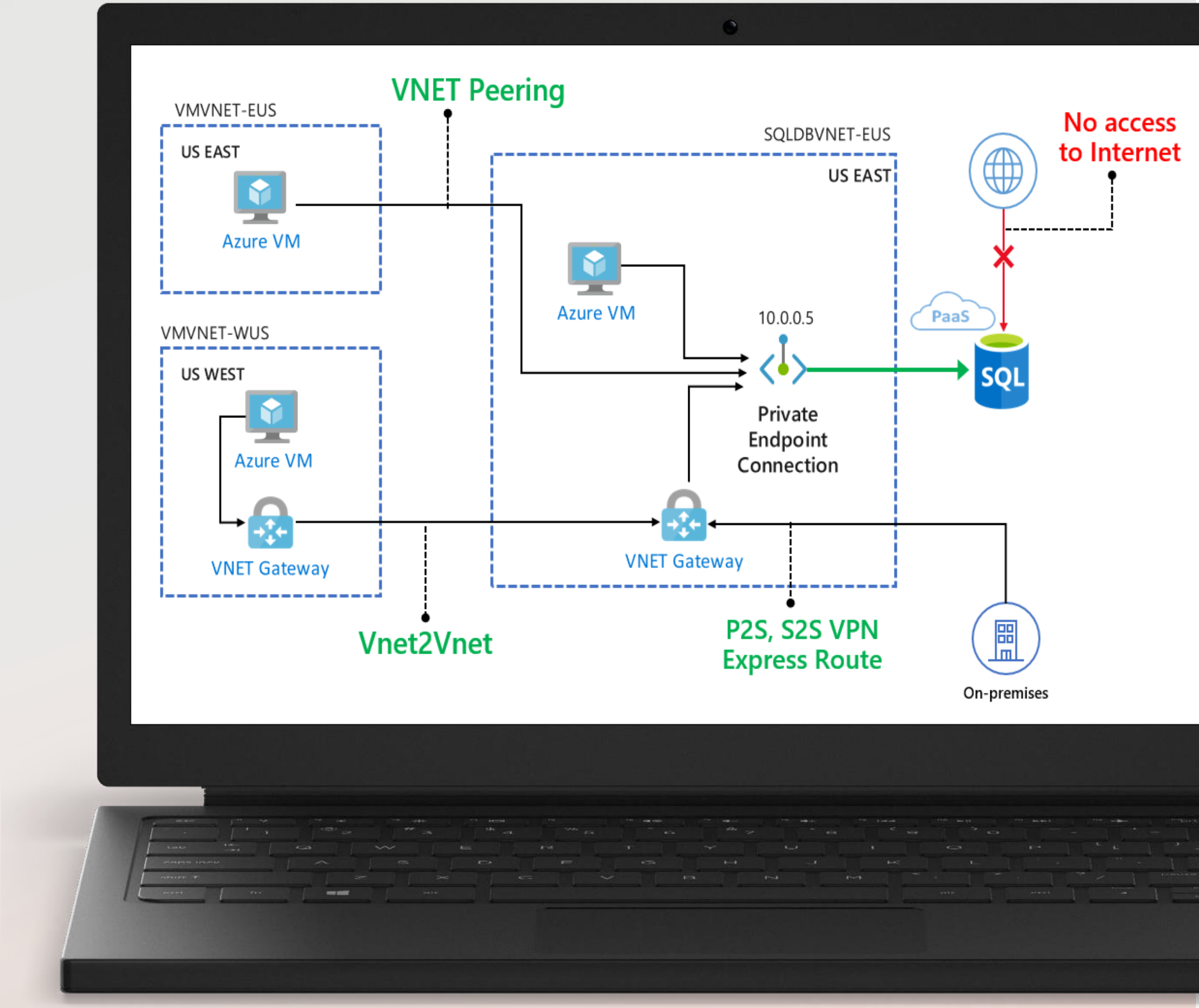
VNET Peering
VMVNET-EUS
US EAST
Azure VM
SQLDBVNET-EUS
US EAST
No access
to Internet
Azure VM
10.0.0.5
PaaS
SQL
Private
Endpoint
Connection
VMVNET-WUS
US WEST
Azure VM
VNET Gateway
VNET Gateway
Vnet2Vnet
P2S, S2S VPN
Express Route
On-premises

Private Endpoint

Attach NIC to VNet

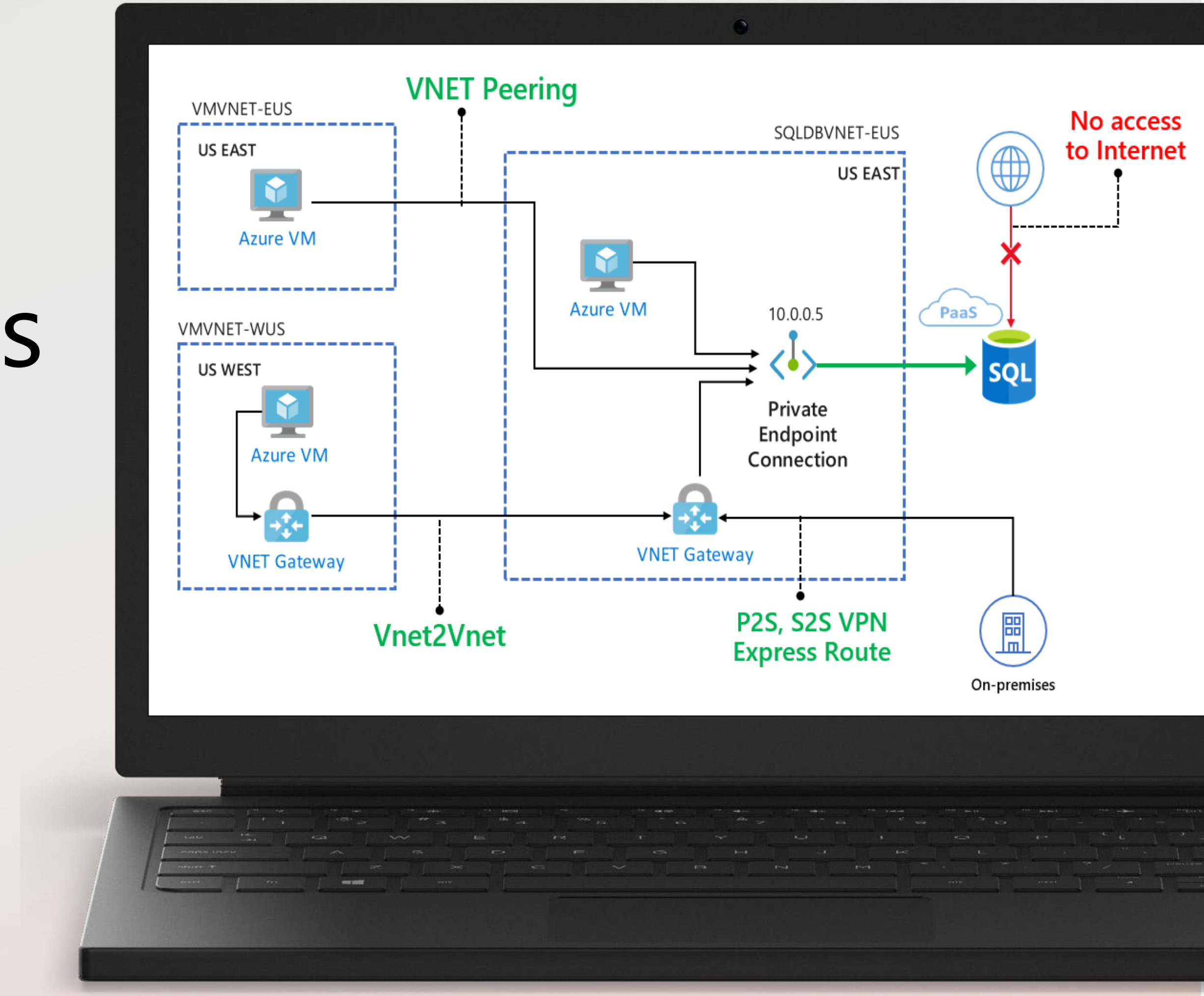
Private DNS-Zone

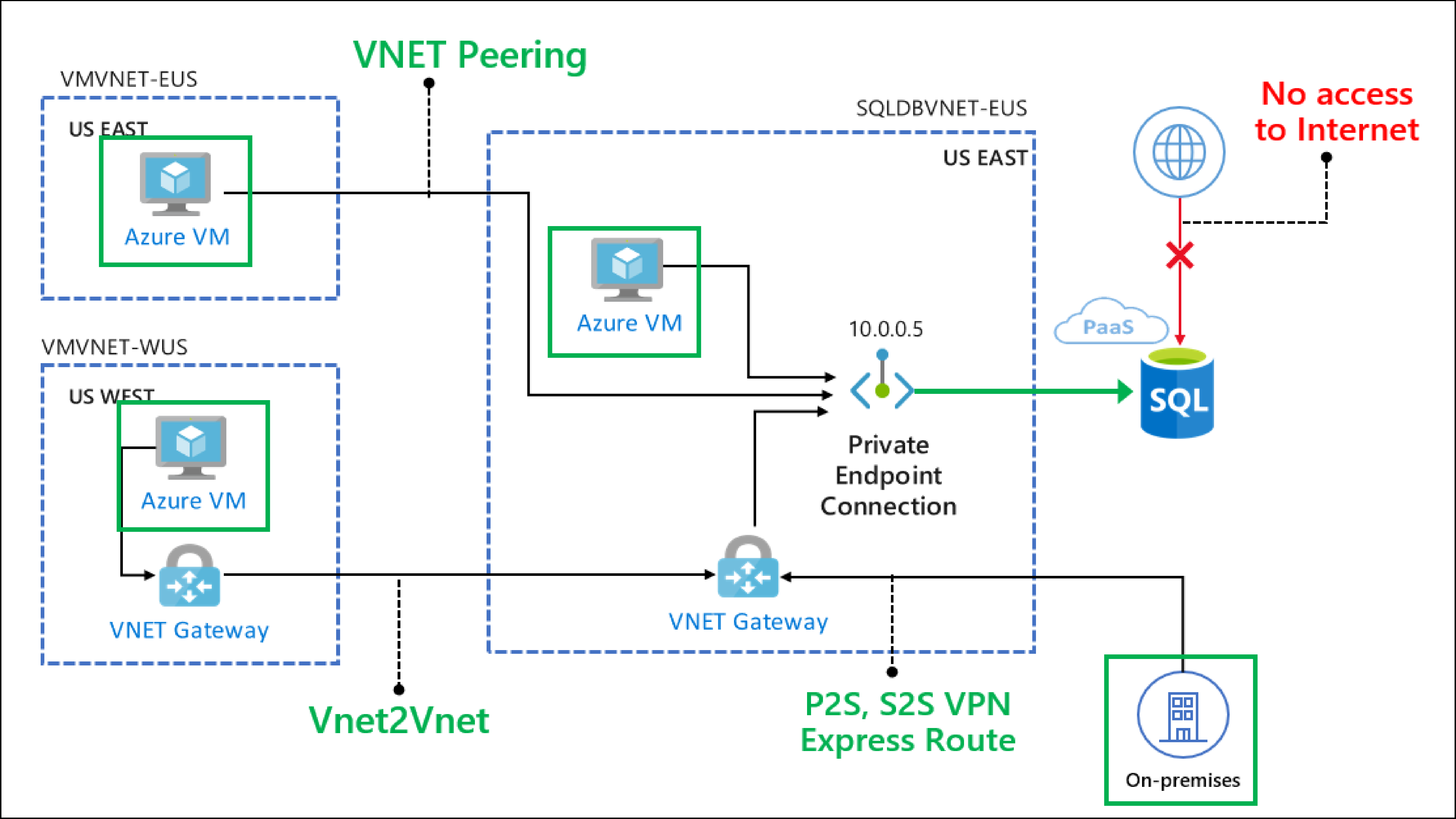
Block public access



Deployments

Public agents don't have access to Private Endpoints





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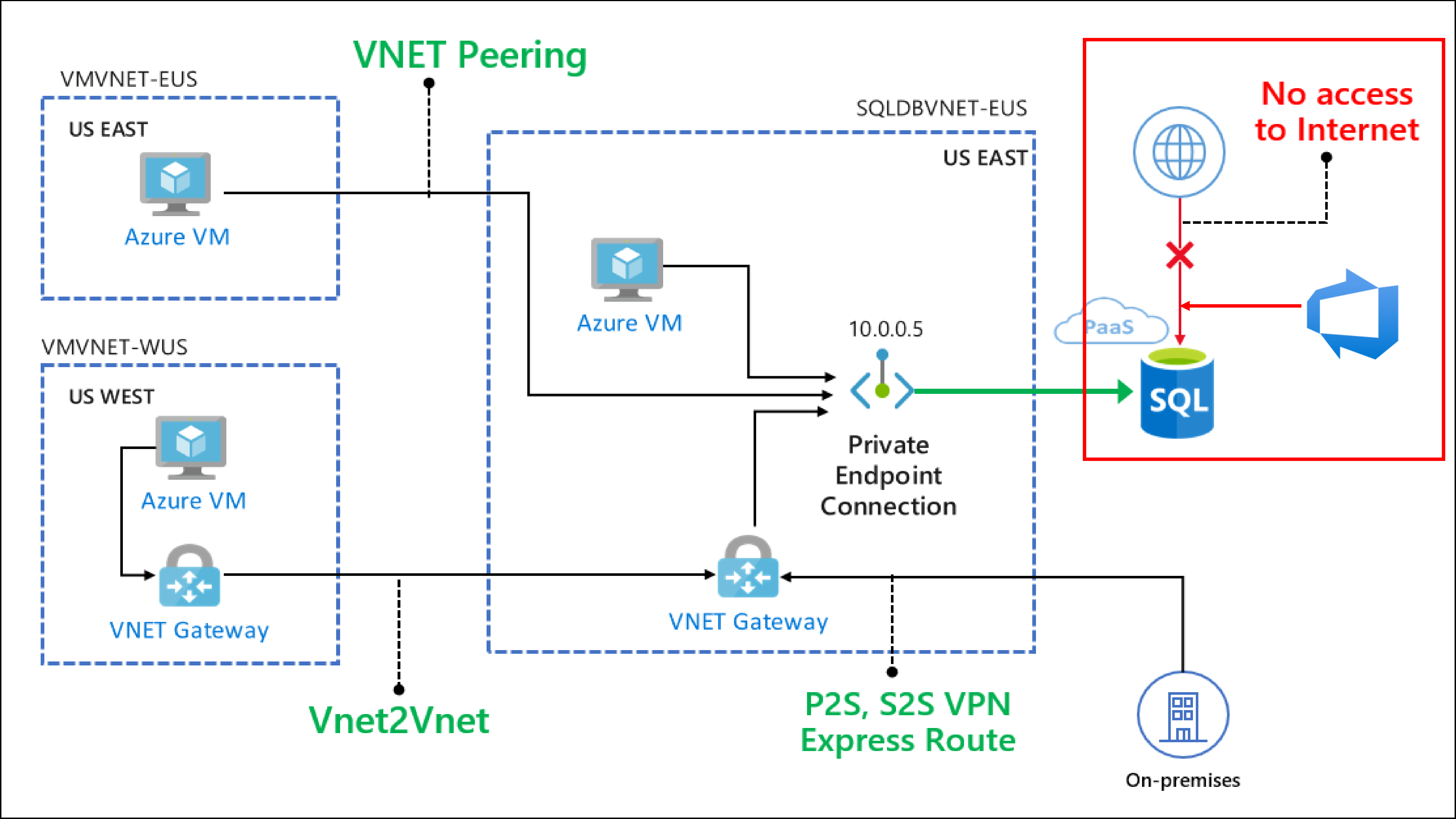
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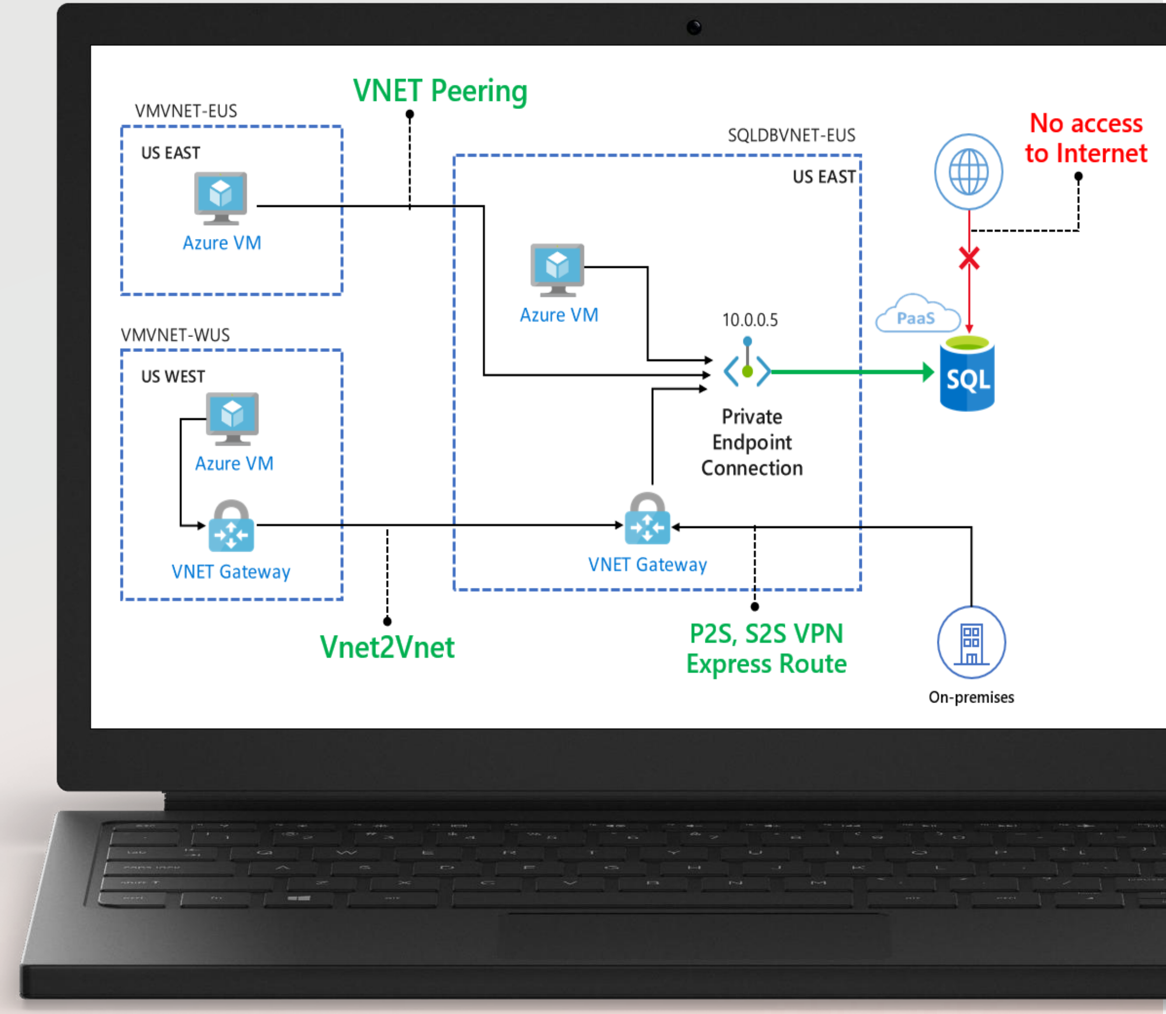
Deployments

Public agents don’t have access to Private Endpoints

Self-hosted agents

Management vs Data Plane



Azure DevOps Self-Hosted Agents

Self-Hosted Agent Advantages

Full control over infrastructure and software

Use specialized agents

Powerful hardware

Self-Hosted Agent Disadvantages

Cost for infrastructure and operations

Monitoring

Maintain OS and software

Self-Hosted Agent Options

Virtual Machine (VMSS)

Run in Container

Managed DevOps Pool

Managed DevOps Pools

Managed DevOps Pools

Best of Microsoft-hosted agent and self-hosted agent on VMs

Azure manages VM and agent

Pay only for usage of VM

Managed DevOps Pools Features

Managed Identity

VNet integration

Custom Image

Stand-by agents

Limited by Azure DevOps licenses

Demo

Demo

Azure Key Vault with Private Endpoint

Deployments only from within VNet

Create Managed DevOps Pool

Test Key Vault access in ADO

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